

ANTIPSYCHOTICS (*Neuroleptics*)

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Psychosis

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- ▶ A **psychiatric disorder** associated with dysfunction of psychological activity with regard to thinking, feeling & behavior.
- ▶ **Schizophrenia:** Principal psychiatric disorder seen in clinical practice
- ▶ Affects about 1-5% of population

Manifestations of Psychosis

❖ Positive symptoms:

- ▶ **Delusions** (*often paranoid in nature*)
- ▶ **Hallucinations** (*perceptions without stimulus; usually in form of vision of things non-existent, hearing voices often with exhortatory or persecutory messages, etc*)
- ▶ **Thought disorder** (*wild trains of thought, distorted sentences and irrational conclusions, sometimes associated with the feeling that thoughts are inserted or withdrawn by an outside agency*)
- ▶ **Abnormal behaviours** (*such as agitation & aggressive behaviours, stereotyped movements, etc*).

❖ **Negative symptoms:**

- ▶ Withdrawal from social contacts, society
- ▶ Flat emotional responses and intellectually dysfunction

Etiology theories of Psychosis

Dopamine theory of Psychosis (by Carlson):

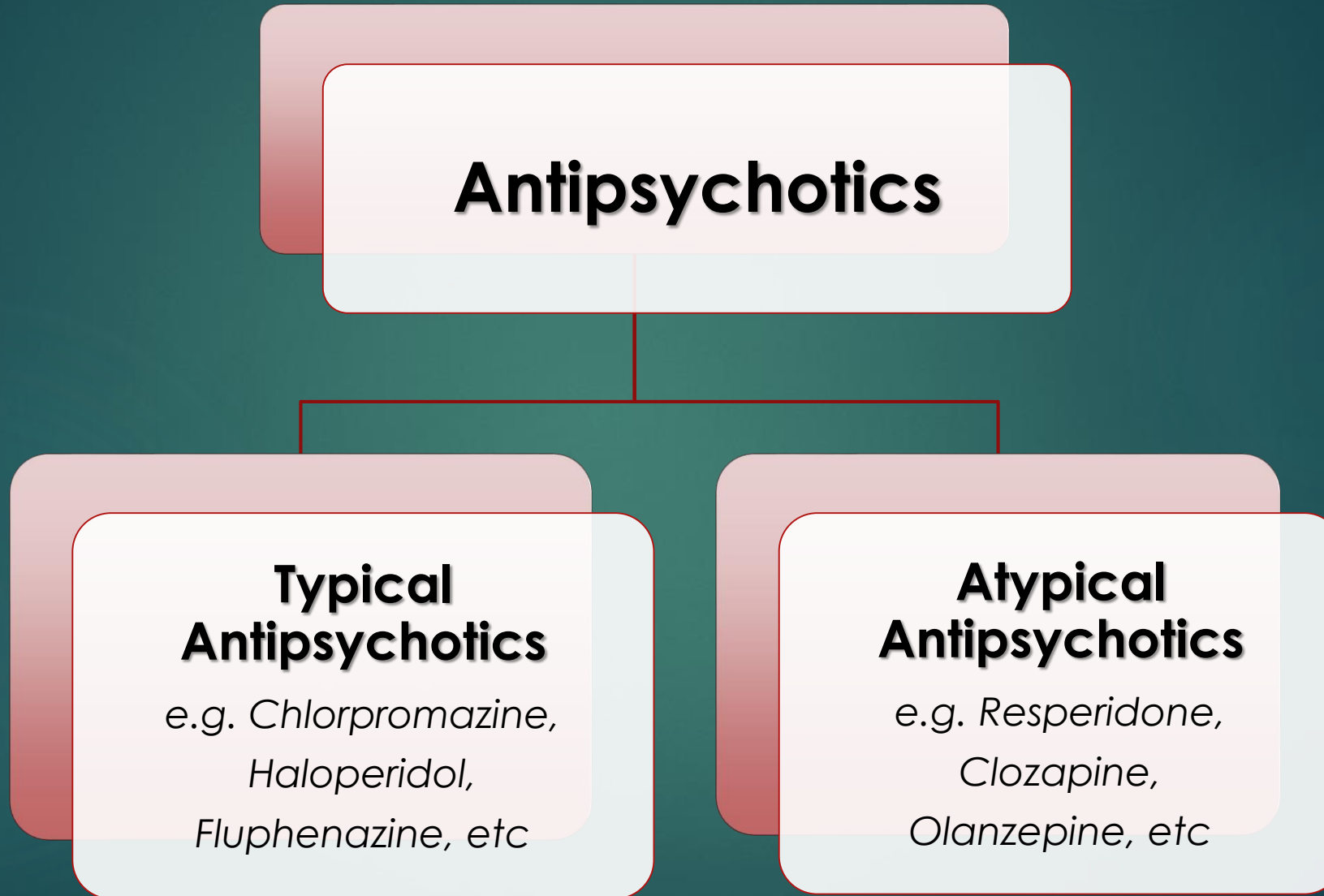
- Psychosis is caused by increased dopaminergic hyperactivity & signal transduction in the brain mesolimbic & mesocortical system
- *Dopamine is the main neurotransmitter activating dopaminergic (D_1 and D_2) receptors in brain mesolimbic system that controls psychological behaviour*
- **Other theories:**
 - Glutamate theory
 - Genetic & Environmental theory

Anti-psychotics

- ▶ Drugs that antagonise/inhibit Dopamine hyperactivity in brain centres
- ▶ Used primarily to treat schizophrenia. (but are also effective in some other psychotic states, e.g. mania and delirium)
- ▶ Antipsychotic drugs are also called *Neuroleptics* or *Antischizophrenics*

Antipsychotic Drug Classification

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Pharmacological Effects of Dopamine D₂ receptor antagonists

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- ▶ Blockade of D₂ receptors in brain mesolimbic system → **Antipsychotic effects**;
- ▶ Blockade of D₂ receptors in brain's chemo-receptor trigger zone (CTZ) → **Anti-emetic effect**;
- ▶ Blockade of D₂ receptors in hypothalamus and pituitary gland → **Gynaecomastia, Galactorrhea, Amenorrhea, Infertility and weight gain** (*side effects*)
- ▶ Blockade of D₂ receptors in brain basal ganglia → **Extrapyramidal side effects** e.g. **Akathisia** (*motor restlessness*), **Dystonia** (*Muscle spasm*), and **Parkinsonism**

N.B. Anticholinergic drug **Benzhexol** given routinely to counteract extrapyramidal side effects)

Antipsychotic drugs (summary)

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Drug Name	Clinical Uses	Adverse Effects
Typical Antipsychotics		
Chlorpromazine	•Psychotic disorders; •CNS-induced nausea & vomiting	<i>Hypotension, tardive dyskinesia, photosensitivity, extrapyramidal side effects, Drowsiness, etc</i>
Haloperidol	•Psychotic disorders; •Tourette's syndrome	<i>Extrapyramidal side effects, akathisia, dystonia, tardive dyskinesia, Hypotension, Drowsiness, etc</i>
Fluphenazine	Psychotic disorders	<i>Extrapyramidal side effects, drowsiness, hypotension, dystonia, akathisia, etc</i>

Antipsychotic drugs (summary)

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Drug Name	Clinical Uses	Adverse Effects
Atypical Antipsychotics		
Resperidone	•Psychotic disorders	Nervousness, akathisia, Agitation, dizziness, constipation, weight gain, pyrexia, etc
Clozapine	•Refractory schizophrenia	Akathisia, seizures, drowsiness, sedation, tachycardia, hypotension, etc
Olanzapine	•Schizophrenia, •Manic phase of bipolar disorder	Nervousness, akathisia, drowsiness, dizziness, constipation, weight gain, etc

Other Antipsychotics: *Thiothixene, Trifluoperazine, Lithium, Prochlorperazine, Quetiapine, Ziprasidone, Trifluoperazine, Pimozide, Loxapine, etc*

Contraindications of Antipsychotics:

- Known hypersensitivity
- Unconscious (comatose) patients
- Severely depressed patients
- Patients with Eosinophilia, Blood dyscrasias or Bone marrow depression
- Parkinson's disease
- Liver impairment
- Coronary artery disease, severe hypotension or hypertension
- Lithium is contraindicated in patients hypersensitive to Tartrazine; Renal or CVD, Natriuresis, Dehydration, Diuretic therapy

Drug Interactions with Antipsychotics

- ▶ Antipsychotic + Alcohol → additive CNS depression
- ▶ Anticholinergics reduce the therapeutic effects of Antipsychotics → worsening of psychotic symptoms and increases tardive dyskinesia
- ▶ Clozapine synergistic with other drugs that suppress bone marrow
- ▶ Drugs that reduce renal clearance increase risk of Lithium toxicity

END